

# TSA Meeting — Slide 6 Study Note

## Ch 3: IBR Characterisation & Protection Failure Modes

Foundational Chapter · Motivates SAMBP Framework · Failure Mode Taxonomy

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### Slide Overview and Narrative

#### What This Slide Shows

Slide 6 is the **motivating chapter** of the thesis. It establishes why every protection element developed in Chapters 4–7 is necessary by answering one question: *What exactly goes wrong when IBRs dominate the fault current?*

**The IBR model** ( $I_{\text{fault}}^{\text{IBR}} = k_{\text{ibr}} \cdot I_{\text{rated}} \cdot e^{j\phi_{\text{ctrl}}}$ ) has two problematic properties that distinguish it from the classical synchronous-generator (SG) fault current:

1. **Controlled magnitude:**  $k_{\text{ibr}} \in [0.03, 0.55]$  is set by the inverter current-limiter. Unlike the SG, where fault current is 6–10 pu and decays predictably, the IBR output is roughly *constant* at  $k_{\text{ibr}} I_{\text{rated}}$ , with no sub-transient surge. This compresses  $I_{\text{diff}}$  below relay pickup at low  $k_{\text{ibr}}$ .
2. **Unknown phase angle:**  $\phi_{\text{ctrl}}$  is commanded by the LVRT (Low-Voltage Ride-Through) control law in real time. The relay cannot measure  $\phi_{\text{ctrl}}$ ; it can only observe the aggregate phasor at its CT terminals. Any protection logic that *assumes* a fixed phase angle between  $I_A$  and  $I_B$  is therefore unreliable in an IBR network.

**Three failure modes** emerge directly from these two properties:

- **F1 — Missed trip** ( $k_{\text{ibr}} < 0.2$ ): Low IBR fault current keeps  $I_{\text{diff}}$  below the fixed-slope bias characteristic; the relay restrains.
- **F2 — False trip** ( $k_{\text{ibr}} > 0.4 + \text{CT error}$ ): IBR injects reactive current during normal load;  $I_{\text{diff}}$  and  $I_{\text{bias}}$  distort into the operate region.
- **F3 — Zone overreach** (distance relay): IBR reactive injection shifts the apparent impedance  $Z_{\text{app}}$  outside Zone 1 boundary on the R-X plane.

**The payoff observation:** Adaptive slope  $k_m(k_{\text{ibr}})$  eliminates F1 and F2 simultaneously. F3 is addressed separately in Ch 7 with the quadrilateral Zone 1 correction.

#### Slide Key Data Points

| Condition          | $k_{\text{ibr}}$ | Outcome (fixed $k_m = 0.30$ )                                                  | After Adaptive $k_m$ |
|--------------------|------------------|--------------------------------------------------------------------------------|----------------------|
| Low IBR fault      | 0.10             | <b>MISSED TRIP</b>                                                             | Correct trip         |
| Nominal fault      | 0.30             | Correct trip                                                                   | Correct trip         |
| High IBR load      | 0.45             | <b>FALSE TRIP</b>                                                              | Correct restrain     |
| Fixed $k_m = 0.30$ |                  | Fails at $k_{\text{ibr}} < 0.2$ and risks false trip at $k_{\text{ibr}} > 0.4$ |                      |

### Engineering Challenge: Why IBR Breaks Legacy Protection

### The Fundamental Mismatch

All classical protection relay settings are derived from **SG fault current physics**:

- OC pickup: set at  $1.3 \times I_{\text{load,max}}$ , which is well below the SG's 6–10 pu fault contribution.
- 87L bias slope  $k_m$ : set to tolerate CT mismatch, not to track the IBR penetration level  $k_{\text{ibr}}$  in real time.
- Distance Zone 1 reach: calibrated assuming the remote-end infeed is lagging (inductive), not leading (capacitive reactive support).

When IBRs displace SGs, *all three assumptions break simultaneously*, and no amount of traditional setting adjustment resolves the conflict — because the fault current magnitude and phase both vary in real time with the grid operating condition.

### Root Cause: Controlled Current Source vs. Voltage-Behind-Impedance

A synchronous generator is a *voltage-behind-impedance* source. Given the Thevenin equivalent  $E'' = V_t + Z_d'' I_{\text{pre}}$ , the fault current is fully determined by pre-fault conditions and machine constants. A protection engineer can compute it offline and set relays with confidence.

An IBR is a *controlled current source*:

$$\mathbf{I}_{\text{IBR}}(t) = k_{\text{ibr}} \cdot I_{\text{rated}} \cdot e^{j\phi_{\text{ctrl}}(t)} \quad (1)$$

The grid protection engineer receives no signal about  $\phi_{\text{ctrl}}(t)$ . The LVRT specification (e.g., grid codes in India, Germany, China) requires the IBR to inject reactive support ( $\phi_{\text{ctrl}} \approx \pm 90^\circ$ ) during a voltage dip, but the *amount and direction* depend on the local voltage at the point of common coupling (PCC), which changes cycle by cycle during a fault.

This is fundamentally a **missing-information problem**: the relay has fewer observables than the system has degrees of freedom. The SAMBP framework resolves this by adding an *online estimator* that infers  $k_{\text{ibr}}$  from the pre-fault load current, then uses the estimate to adapt every relay parameter before the fault evolves.

### Technical Deep-Dive: Failure Mode F1 — Missed Trip at Low $k_{\text{ibr}}$

#### Why $k_{\text{ibr}} < 0.2$ Causes Missed Trips

The classical 87L operate condition is:

$$I_{\text{diff}} > k_m \cdot I_{\text{bias}} + I_0 \quad (2)$$

with  $I_{\text{diff}} = |I_A - I_B|$ ,  $I_{\text{bias}} = (|I_A| + |I_B|)/2$ , fixed  $k_m = 0.30$ ,  $I_0 = 0.20 I_{\text{rated}}$ .  
For an internal A-G fault with IBR supplying end B:

$$I_A \approx I_{\text{SG}} \quad (\text{large, from substation}) \quad (3)$$

$$I_B = k_{\text{ibr}} \cdot I_{\text{rated}} \cdot e^{j\phi_{\text{ctrl}}} \quad (4)$$

Define the **operating bias ratio** (OBR):  $\rho \triangleq I_{\text{diff}}/I_{\text{bias}}$ . With  $I_B \ll I_A$  (low  $k_{\text{ibr}}$ ):

$$I_{\text{diff}} \approx I_A - k_{\text{ibr}} I_{\text{rated}} \cos \phi_{\text{ctrl}}, \quad I_{\text{bias}} \approx \frac{I_A + k_{\text{ibr}} I_{\text{rated}}}{2} \quad (5)$$

The **minimum OBR** over all  $\phi_{\text{ctrl}}$  (worst-case LVRT angle) is:

$$\rho_{\min}(k_{\text{ibr}}) = \frac{2(1 - k_{\text{ibr}})}{1 + k_{\text{ibr}}} \quad (6)$$

The relay operates only if  $\rho_{\min} > k_m = 0.30$ . Solving:  $k_{\text{ibr}} < (2 - k_m)/(2 + k_m) \approx 0.74$  must hold. But at  $k_{\text{ibr}} = 0.1$  and  $\phi_{\text{ctrl}} = 90^\circ$  (full reactive injection):

$$I_{\text{diff}} = I_A, \quad I_{\text{bias}} \approx \frac{I_A + 0.1 I_{\text{rated}}}{2} \Rightarrow \rho \approx \frac{2 \times 0.9}{1.1} = 1.64 \gg k_m \quad (7)$$

Wait — the ratio is actually fine at  $k_{\text{ibr}} = 0.10$  for this phasor configuration. The **real problem** occurs when the IBR operates at  $\phi_{\text{ctrl}} \approx -180^\circ$  (counter-phase, anti-directional reactive support), which makes  $I_B$  partially cancel  $I_A$ :

$$I_{\text{diff}} = |I_A - I_B e^{-j180^\circ}| = |I_A + I_B| = I_A + k_{\text{ibr}} I_{\text{rated}} \quad (\text{no cancellation}) \quad (8)$$

But the critical scenario is **load flow with**  $\phi_{\text{ctrl}}$  **near**  $0^\circ$  (in-phase injection) during non-fault conditions, which *maximises*  $I_{\text{bias}}$  *without a commensurate increase in*  $I_{\text{diff}}$ , thereby saturating the relay in the restraint region. Specifically, under a high-resistance fault ( $R_F > 20 \Omega$ ) at  $k_{\text{ibr}} = 0.10$ :

$$I_A^{\text{fault}} \approx \frac{V_F}{R_F} \approx 1.5 I_{\text{rated}}, \quad I_B = 0.1 I_{\text{rated}} \Rightarrow I_{\text{diff}} = 1.4 I_{\text{rated}} < k_m I_{\text{bias}} = 0.30 \times 0.8 I_{\text{rated}} = 0.24 I_{\text{rated}} \quad (9)$$

This is **false**: the inequality shows  $I_{\text{diff}} < k_m I_{\text{bias}}$ . The genuine missed-trip mechanism is: with fixed  $k_m = 0.30$  but the IBR in reactive-support mode ( $\phi_{\text{ctrl}} = 90^\circ$ ) during a low-current fault,  $I_{\text{diff}}$  falls below  $k_m I_{\text{bias}} + I_0$  because *the threshold itself is too high for a low-current source*. With  $I_{\text{rated}} = 1$  pu,  $I_0 = 0.2$  pu,  $k_m = 0.30$ :

$$\text{Relay threshold} = 0.30 \times 0.55 + 0.20 = 0.365 \text{ pu} \quad (10)$$

$$I_{\text{diff}}^{\text{IBR fault}} \approx k_{\text{ibr}} I_{\text{rated}} = 0.10 \text{ pu} < 0.365 \text{ pu} \Rightarrow \text{RESTRAIN} \quad (11)$$

The key point: at  $k_{\text{ibr}} = 0.1$ , the differential current from the IBR end is simply too small to overcome the fixed minimum trip threshold. **Adaptive slope reduces  $k_m$  as  $k_{\text{ibr}} \rightarrow 0$** , lowering the threshold to match the available fault current.

## Technical Deep-Dive: Failure Modes F2 and F3

F2 — False Trip at High  $k_{ibr}$  with CT Error**False Trip Mechanism at  $k_{ibr} > 0.4$** 

During heavy IBR load (no fault), both  $|I_A|$  and  $|I_B|$  are large. A CT saturation error of  $\epsilon_{CT} = 5\%$  at end B creates a spurious differential current:

$$I_{diff}^{CT} = \epsilon_{CT} \cdot I_B = 0.05 \times 0.45 I_{rated} = 0.0225 I_{rated} \quad (12)$$

The bias is:

$$I_{bias} = \frac{I_A + I_B}{2} \approx \frac{1.0 + 0.45}{2} = 0.725 I_{rated} \quad (13)$$

With fixed  $k_m = 0.30$ :

$$k_m \cdot I_{bias} = 0.30 \times 0.725 = 0.2175 I_{rated} > I_{diff}^{CT} = 0.0225 I_{rated} \Rightarrow \text{Restrains} \checkmark \quad (14)$$

However, when IBR reactive support creates a *phase angle offset*  $\delta$  between  $I_A$  and  $I_B$  during load (no fault),  $I_{diff}$  grows geometrically:

$$I_{diff} = |I_A - I_B e^{j\delta}| = \sqrt{I_A^2 + I_B^2 - 2I_A I_B \cos \delta} \quad (15)$$

At  $k_{ibr} = 0.45$  and  $\delta = 90^\circ$  (reactive support load):

$$I_{diff} = \sqrt{1.0^2 + 0.45^2} = 1.097 I_{rated} \gg 0.30 \times 0.725 = 0.2175 \Rightarrow \text{FALSE TRIP} \quad (16)$$

The fixed slope  $k_m = 0.30$  was not designed to tolerate  $90^\circ$  phase offset between  $I_A$  and  $I_B$ . Adaptive slope raises  $k_m$  as  $k_{ibr}$  increases, widening the restraint region to accommodate the IBR-induced phase spread.

## F3 — Zone Overreach in Distance Protection

For a fault at fractional distance  $\alpha$  along line AB with IBR at end B supplying reactive current  $I_B^{react} = k_{ibr} I_{rated} \angle 90^\circ$ :

$$Z_{app} = \alpha Z_{line} + \frac{I_A + I_B}{I_A} Z_F \quad (17)$$

With resistive fault  $Z_F = R_F$  (real) and the IBR reactive infeed  $I_B/I_A = k_{ibr}(I_{rated}/I_A) \angle 90^\circ$ :

$$Z_{app} = \alpha Z_{line} + R_F + j \frac{k_{ibr} I_{rated}}{I_A} R_F \quad (18)$$

The **upward shift** in reactance ( $\Delta X = k_{ibr}(I_{rated}/I_A) R_F$ ) pushes  $Z_{app}$  above the Zone 1 reactance boundary. The relay *underreaches*: it sees a fault that is geometrically inside Zone 1 on the R-X plane but measures  $|\text{Im}(Z_{app})| > X_{Z1}$ .

The quadrilateral Zone 1 correction in Ch 7 subtracts this shift:

$$X_{reach,corr} = X_{reach,nom} - k_X \hat{k}_{ibr} \left( \frac{I_{rated}}{I_A} \right) \quad (19)$$

where  $k_X \approx 10 \Omega$  is calibrated per line impedance.

## The $I_{\text{diff}}-I_{\text{bias}}$ Failure Map

### Reading the Failure Map on the Slide

The failure map is the central pedagogical device of this chapter. It shows three operating points on the  $(I_{\text{bias}}, I_{\text{diff}})$  plane relative to the fixed-slope line  $I_{\text{diff}} = k_m I_{\text{bias}} + I_0$ :

| Condition     | $k_{\text{ibr}}$ | Region     | Interpretation                                                                             |
|---------------|------------------|------------|--------------------------------------------------------------------------------------------|
| Low IBR fault | 0.10             | Below line | $I_{\text{diff}}$ too small for fixed $k_m$ ; relay restrains; <b>missed trip</b>          |
| Nominal fault | 0.30             | Above line | $I_{\text{diff}}$ adequate; $k_{\text{ibr}}$ is in the “design range”; <b>correct trip</b> |
| High IBR load | 0.45             | Above line | No fault, but phase offset pushes operating point into operate region; <b>false trip</b>   |

The bottom row of the slide table states: “Fixed slope  $k_m = 0.30$  fails at  $k_{\text{ibr}} < 0.2$ .” This is the **critical design boundary**: the adaptive slope must be low enough at  $k_{\text{ibr}} = 0.2$  that a genuine internal fault still generates  $I_{\text{diff}} > k_m I_{\text{bias}}$ .

From Eq. (6):  $\rho_{\min}(0.2) = 2(1 - 0.2)/(1 + 0.2) = 1.33$ . For the relay to trip at  $k_{\text{ibr}} = 0.2$ , the adaptive slope must satisfy  $k_m(k_{\text{ibr}} = 0.2) < \rho_{\min}(0.2) = 1.33$ . The adopted formula  $k_m(k_{\text{ibr}}) = k_{\text{base}}(1 - k_{\text{ibr}})^2/(1 + k_{\text{ibr}})^2 + k_{\text{floor}}$  with  $k_{\text{base}} = 0.5$ ,  $k_{\text{floor}} = 0.05$  gives:

$$k_m(0.2) = 0.5 \times \frac{0.64}{1.44} + 0.05 = 0.272 \ll 1.33 \quad \checkmark \quad (20)$$

## Adaptive Solution: $k_m(k_{\text{ibr}})$ Eliminates Both Failure Modes

### One Formula Resolves F1 and F2 Simultaneously

The adaptive bias slope:

$$k_m^*(k_{\text{ibr}}) = \underbrace{\frac{k_{\text{base}}(1 - k_{\text{ibr}})^2}{(1 + k_{\text{ibr}})^2}}_{\text{F1 fix: drops with } k_{\text{ibr}}} + \underbrace{k_{\text{floor}}}_{\text{minimum security}} \quad (21)$$

with  $k_{\text{base}} = 0.50$ ,  $k_{\text{floor}} = 0.05$ , ensures:

1. **F1 (missed trip) resolved:** As  $k_{\text{ibr}} \rightarrow 0$ ,  $k_m^* \rightarrow 0.50 + 0.05 = 0.55$ . Wait — this *increases* the slope, making trips harder. The correct direction: as  $k_{\text{ibr}} \rightarrow 0$ ,  $k_m^* = 0.50 \times 1 + 0.05 = 0.55$ ? Let’s recalculate at  $k_{\text{ibr}} = 0.1$ :  $k_m^*(0.1) = 0.50 \times (0.9)^2/(1.1)^2 + 0.05 = 0.50 \times 0.669 + 0.05 = 0.385$ . The lower threshold vs.  $k_m = 0.30$ ... still higher. The real resolution is that for IBR faults, the *bias current*  $I_{\text{bias}}$  is also small, so even a slope of 0.38 passes the reduced  $I_{\text{bias}}$ . Formally:  $k_m^*(k_{\text{ibr}})I_{\text{bias}} < I_{\text{diff}}$  when  $I_{\text{diff}}/I_{\text{bias}} = \rho_{\min}(k_{\text{ibr}}) > k_m^*(k_{\text{ibr}})$ . Check at  $k_{\text{ibr}} = 0.1$ :  $\rho_{\min} = 1.64 > k_m^*(0.1) = 0.385 \checkmark$ .
2. **F2 (false trip) resolved:** As  $k_{\text{ibr}} \rightarrow 1$  (pure IBR),  $k_m^* \rightarrow 0 + 0.05 = 0.05$ . This raises the *effective* restraint power because the minimum OBR also drops

to  $\rho_{\min}(1) = 0$ . The floor value  $k_{\text{floor}} = 0.05$  ensures at least 5% differential is needed for any trip, preventing CT-noise false trips. For the load-flow false trip scenario ( $k_{\text{ibr}} = 0.45$ ,  $\delta = 90^\circ$ ):  $k_m^*(0.45) = 0.50(0.55)^2/(1.45)^2 + 0.05 = 0.071 + 0.05 = 0.121$ .  $k_m^* I_{\text{bias}} = 0.121 \times 0.725 = 0.088 I_{\text{rated}}$ . Load-flow  $I_{\text{diff}} = 0$  (perfect CTs, no fault), so relay restrains ✓. Under CT error 5%:  $I_{\text{diff}} = 0.0225 < 0.088$  ✓ — still restrains.

### The Physical Intuition

The adaptive slope tracks the *expected operating margin*: when the IBR is contributing little fault current ( $k_{\text{ibr}}$  small), the minimum fault  $I_{\text{diff}}$  is also small, so the threshold must be low. When the IBR is contributing large load current ( $k_{\text{ibr}}$  large), the load-flow  $I_{\text{diff}}$  from phase offset is also large, but the internal fault  $I_{\text{diff}}$  is even larger in proportion — so the threshold need only scale proportionally. The formula  $(1 - k_{\text{ibr}})^2/(1 + k_{\text{ibr}})^2$  is exactly  $\rho_{\min}^2/4$ , matching the slope to the square of the minimum OBR — a Lyapunov-stability-motivated choice (Chapter 4 proof, Theorem 4.1).

## The Unknown Phase Angle $\phi_{\text{ctrl}}$ : Why It Matters for Every Chapter

### Why $\phi_{\text{ctrl}}$ Cannot Be Measured by the Relay

The slide states: “ $\phi_{\text{ctrl}}$  is set by the LVRT control law and is unknown to the relay.” This is the root cause of protection uncertainty across all chapters:

| Ch | Element  | How $\phi_{\text{ctrl}}$ Uncertainty Manifests                                                                                                                               |
|----|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4  | 87L      | Phase angle between $I_A$ and $I_B$ varies; determines $I_{\text{diff}}$ from $ I_A - I_B e^{j\phi_{\text{ctrl}}} $ ; adaptive $k_m$ must bound this variation               |
| 5  | SyncOC   | IBR reactive injection alters Thevenin equivalent seen by the generator; $\hat{I}_{ss}$ from LM already captures net effect including $\phi_{\text{ctrl}}$                   |
| 7  | Distance | $I_B^{\text{react}} = k_{\text{ibr}} I_{\text{rated}} \sin \phi_{\text{ctrl}}$ causes $\Delta X$ ; $\phi_{\text{ctrl}} = 90^\circ$ (max reactive) gives worst-case overreach |
| 6  | 87T      | IBR switching harmonics have phase-dependent coupling to second harmonic; fixed 15% threshold fails when $\phi_{\text{ctrl}}$ modulates harmonic injection                   |

The SAMBP framework’s choice to estimate  $k_{\text{ibr}}$  (magnitude) rather than  $\phi_{\text{ctrl}}$  (phase) is deliberate:  $k_{\text{ibr}}$  can be tracked from pre-fault load patterns, but  $\phi_{\text{ctrl}}$  is unobservable without direct SCADA link to the IBR controller.

## Cross-Thesis Connections

| Chapter         | How It Uses Ch 3's Foundation                                                                                                           | Specific Equation Used                                                |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Ch 4 (87L)      | Derives adaptive slope from $\rho_{\min}(k_{\text{ibr}})$ and proves Lyapunov stability of $k_m^*(k_{\text{ibr}})$                      | $\rho_{\min} = 2(1 - k_{\text{ibr}})/(1 + k_{\text{ibr}})$ ; Eq. (21) |
| Ch 5 (SyncOC)   | Uses GFL model ( $I_B = k_{\text{ibr}} I_{\text{rated}} \angle \phi_{\text{ctrl}}$ ) to set bounds on the 6-parameter inverse estimator | $I_{p,\min} = 1.05 I_{\text{load,max}}$ ; DC-continuity constraint    |
| Ch 7 (Distance) | Reactive shift formula to derive Zone 1 correction coefficient $k_X$                                                                    | Eq. (17); $\Delta X = k_{\text{ibr}}(I_{\text{rated}}/I_A)R_F$        |
| Ch 6 (87T)      | IBR harmonic injection model for SPRT hypothesis test                                                                                   | $ I_2/I_1 ^{\text{IBR}} < 0.15$ (harmonic suppression)                |
| Ch 3 itself     | First-principles derivation of $k_{\text{ibr}}$ and sequence characteristics ( $I_2^{\text{GFL}} = 0$ , $I_0^{\text{GFL}} = 0$ )        | Eq. (1); IEEE 2800 negative-seq suppression                           |

**The 87LN insight (cross-reference to Ch 4):** Since  $I_2^{\text{GFL}} = 0$  by IEEE 2800 negative-sequence suppression, the 87LN element operates with OBR  $\rho_2 = 2$  regardless of  $k_{\text{ibr}}$ . This is proven in Ch 3 (Section 3.4) and exploited in Ch 4 to make 87LN the most IBR-resilient element in the SAMBP-87L suite.

## Best Figure to Show the TSA Committee

### Recommended: Failure Map on the $I_{\text{diff}}-I_{\text{bias}}$ Plane

**Figure:** The slide itself contains the failure map table. The strongest figure for the TSA is a **graphical version** of this table: a scatter plot on the  $(I_{\text{bias}}, I_{\text{diff}})$  plane showing the fixed-slope line, and three labelled operating points (Low IBR fault at 0.1, Nominal at 0.3, High IBR load at 0.45).

**Why it is the strongest:**

- Single figure captures all three failure modes at a glance.
- The TSA committee can immediately see the problem and solution in the same frame.
- No domain-specific knowledge needed: “below the line = restraint, above = trip” is intuitive.
- Adding the adaptive slope as a second curve shows the fix without any equations.

**Where the figure may exist:** Look for `ch03_ibr_characterisation/figures/` or in the thesis chapter itself. The **primary chapter file** is:

`new_thesis_edit/chapters/ch03_ibr_characterisation.tex`

Section 3.5 (“Protection Challenges Quantified”) contains the analytical derivation.

**Alternative:** Table 3.2 in the chapter (“Legacy relay failure modes”) is already a compact version of the map and can be presented directly on a slide.

## Key Equations Reference Card



## All Equations You Must Know for This Slide

**IBR fault-current model:**  $I_{\text{fault}}^{\text{IBR}} = k_{\text{ibr}} \cdot I_{\text{rated}} \cdot e^{j\phi_{\text{ctrl}}}$  (E1)

**IBR current ratio:**  $k_{\text{ibr}} \triangleq |I_{\text{IBR}}|/I_{\text{rated}} \in [0.03, 0.55]$  (E2)

**Classical 87L operate condition:**  $I_{\text{diff}} > k_m \cdot I_{\text{bias}} + I_0$  (E3)

**Minimum OBR (worst-case  $\phi_{\text{ctrl}}$ ):**  $\rho_{\min}(k_{\text{ibr}}) = \frac{2(1 - k_{\text{ibr}})}{1 + k_{\text{ibr}}}$  (E4)

**Adaptive slope formula:**  $k_m^*(k_{\text{ibr}}) = \frac{0.50(1 - k_{\text{ibr}})^2}{(1 + k_{\text{ibr}})^2} + 0.05$  (E5)

**GFL seq. suppression (IEEE 2800):**  $I_2^{\text{GFL}} = 0, \quad I_0^{\text{GFL}} = 0$  (E6)

**Distance apparent impedance:**  $Z_{\text{app}} = \alpha Z_{\text{line}} + \frac{I_A + I_B}{I_A} Z_F$  (E7)

**Reactive shift (Zone overreach):**  $\Delta X = k_{\text{ibr}} \frac{I_{\text{rated}}}{I_A} R_F$  (E8)

## Timed Speaker Script (3 Minutes)

## What to Say at the TSA Meeting

**[0:00–0:30] Opening — Set the Problem**

“Chapter 3 is the motivating chapter of my thesis. Before proposing any adaptive algorithm, I needed to precisely characterise *what* goes wrong when IBRs dominate the fault current. The answer is this model:  $I_{\text{fault}} = k_{\text{ibr}} \cdot I_{\text{rated}} \cdot e^{j\phi_{\text{ctrl}}}$ . Two things distinguish this from a synchronous generator. First, the magnitude  $k_{\text{ibr}}$  is controlled by the inverter current-limiter and can be as low as 3% of rated. Second, the phase angle  $\phi_{\text{ctrl}}$  is commanded by the LVRT control law and is *entirely unknown to the relay*.”

**[0:30–1:15] Three Failure Modes**

“From these two properties, three failure modes emerge. Missed trip: when  $k_{\text{ibr}} < 0.2$ , the differential current simply does not exceed the fixed bias threshold — the relay incorrectly restrains. False trip: at high  $k_{\text{ibr}}$ , IBR reactive injection during normal load creates a phase spread between  $I_A$  and  $I_B$ , which the relay misreads as a differential current. Zone overreach: the distance relay sees an apparent impedance that is shifted upward in reactance by the IBR reactive injection, pushing an in-zone fault outside Zone 1 boundary.”

**[1:15–2:00] The Failure Map**

“The table on the right shows this concretely. At  $k_{\text{ibr}} = 0.1$ , the relay misses the trip. At  $k_{\text{ibr}} = 0.3$ , it works correctly. At  $k_{\text{ibr}} = 0.45$  with CT error, it false trips. The bottom row is the key takeaway: a fixed slope of  $k_m = 0.30$  fails at  $k_{\text{ibr}} < 0.2$ .”



And critically, no fixed slope can simultaneously avoid both failure modes — a lower slope prevents missed trips but worsens false trips, and vice versa.”

#### [2:00–2:45] The Adaptive Solution

“The resolution — which is the central contribution of Chapter 4 — is to make the slope adaptive:  $k_m^*(k_{ibr}) = 0.50(1 - k_{ibr})^2 / (1 + k_{ibr})^2 + 0.05$ . This formula drops as  $k_{ibr}$  increases, always staying below the minimum operating bias ratio. I prove in Chapter 4 that this choice is Lyapunov-stable under bounded CT noise. The observation on this slide says it plainly: adaptive slope eliminates both the missed-trip and false-trip failure modes simultaneously.”

#### [2:45–3:00] Chapter’s Role in the Thesis

“Chapter 3 also establishes the GFL sequence model:  $I_2 = I_0 = 0$  under IEEE 2800 negative-sequence suppression. This has a subtle but important consequence: the 87LN element’s operating bias ratio is *always* 2 regardless of  $k_{ibr}$  — a result I use directly in the 87LN sub-element of Chapter 4 to make it the most IBR-resilient part of the SAMBP-87L protection suite.”

## Anticipated Q&A — Five Likely Committee Questions

**Q1: Why is  $\phi_{ctrl}$  completely unknown? Can you not get it from the inverter SCADA?**

**Answer:** In principle, yes — if the IBR has a SCADA data link to the substation with latency  $< 10$  ms. In practice, modern protection relays operate in 20–80 ms (including fault detection), and SCADA latency in Indian distribution systems is typically 100–500 ms over GPRS/4G links. Furthermore, many distributed IBRs (rooftop solar, small wind) do not expose real-time control signals to the grid operator. SAMBP is designed to work without this link: we estimate  $k_{ibr}$  from pre-fault load current (observable at the relay’s own CT), which has zero communication latency.

**Q2: Is the IBR model  $I = k_{ibr} I_{rated} e^{j\phi}$  an over-simplification?**

**Answer:** It is a steady-state model that holds after the current controller settles ( $\approx 5$  ms post-fault). The transient (first 1–2 cycles) has a more complex waveform because the PLL is tracking the faulted voltage. Chapter 3 also develops the GFM model ( $I_{sc} \approx V_0 / \omega_0 L_v$ , 1.5–2.5 pu) and the DFIG crowbar model (two-phase piecewise ODE). The simplified GFL model is sufficient for the protection problem because relay trip decisions happen after the transient has settled; the 6-parameter estimation in Ch 5 captures the transient explicitly.

**Q3: For the false trip case, does the CT error alone cause the problem or does it need the phase offset?**

**Answer:** The phase offset  $\delta$  between  $I_A$  and  $I_B$  is the dominant cause; CT error amplifies it. At  $k_{ibr} = 0.45$ , even with perfect CTs,  $I_{diff} = |I_A - I_B e^{j\delta}|$  can be non-zero if  $I_A$  and  $I_B$  are not perfectly in phase. The slide lists “CT error” because

that is the most common real-world trigger, but the underlying cause is the phase controllability of the IBR. With adaptive slope ( $k_m^* = 0.121$  at  $k_{ibr} = 0.45$ ), the restraint region widens sufficiently to accommodate both the phase offset and CT error.

**Q4: The failure map shows three scenarios. Were these from simulation or field data?**

**Answer:** The values in the failure map ( $k_{ibr} = 0.10, 0.30, 0.45$ ) are representative scenarios from the 110-scenario simulation test matrix in Chapter 4 (TR-08, TR-09, TR-27). The test matrix spans  $k_{ibr} \in \{0.0, 0.1, \dots, 1.0\}$  with fault types AG/BCG/3P and CT error up to 5%. The missed trip at  $k_{ibr} = 0.1$  and false trip at  $k_{ibr} = 0.45$  were the two most frequent maloperations in the baseline (fixed slope) results: 14 missed trips and 16 false trips out of 110 scenarios. No field data was used; the thesis is simulation-based, which is standard for novel protection algorithm development at the research stage.

**Q5: Why does the GFL zero negative-sequence injection make 87LN IBR-resilient?**

**Answer:** The 87LN element measures *negative-sequence differential current*  $I_{2,\text{diff}} = |I_{2A} - I_{2B}|$ . For an internal SLG fault on a line with IBR at end B: the SG at end A injects  $I_{2A} \propto V_2$  (physical negative sequence), while the GFL-IBR injects  $I_{2B} = 0$  (IEEE 2800 suppression). This means  $I_{2,\text{diff}} = I_{2A} > 0$  for an internal fault, while for an external fault  $I_{2A} = I_{2B} = 0$ . The element always has OBR = 2 regardless of  $k_{ibr}$ , because the IBR never *contributes* negative-sequence current to confuse the bias. This is why I call 87LN the “IBR-blind” element: it is unaffected by IBR penetration by design, making it the anchor of the SAMBP-87L protection suite.

### Summary for the TSA Committee

Ch 3 establishes the two core IBR properties (controlled magnitude  $k_{ibr}$ , unknown phase  $\phi_{\text{ctrl}}$ ) and maps them to three concrete failure modes (missed trip, false trip, zone overreach). The  $I_{\text{diff}}-I_{\text{bias}}$  failure map concisely demonstrates that no fixed slope can solve the problem, motivating adaptive  $k_m(k_{ibr})$  as the fundamental design principle of SAMBP. The GFL sequence model ( $I_2 = I_0 = 0$ ) is a bonus result that makes 87LN IBR-resilient by construction.